



ArogyaPulse

Problem Statement: *Build technology that improves healthcare accessibility, prevention, wellness, patient experience, medical workflows or healthcare delivery.*

Domain: *HealthTech*

Team: *Hack Zombie*

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HACK
MATRIX

AROGYAPULSE

AI-Powered Rural Healthcare Triage System

Every rural pulse, seen in time.

Arav Kotwal · Agastaya Jindal · Adithya Ranjan · Karthik Natarajan

Madhav Institute of Technology & Science, Gwalior



THE PROBLEM

The Silence Between Heartbeats

01

No Real-Time Tools

ASHA workers lack simple digital tools for quick clinical triage at the village level.

02

Delayed Detection

High-risk cases reach doctors only after emergencies have already begun.

03

Last Mile Unserved

Language barriers and complex apps leave rural communities without support.

WHY THIS PROBLEM EXISTS

- Cause 1** No digital triage tools at village level
- Cause 2** High patient load + shortage of doctors
- Cause 3** Paper registers create dangerous delays

CORE PROBLEM

No instant risk stratification

USER CONSEQUENCE

Late referrals & missed red flags

REAL IMPACT

Preventable maternal & emergency deaths

Current Approach: Paper Register → Manual Guesswork → Delayed Review → Emergency

OUR SOLUTION

Introducing ArogyaPulse

ArogyaPulse enables ASHA workers to instantly risk-stratify patients and pushes critical cases to doctors in real time.

ASHA → Vitals → Engine → Risk + AI → Doctor

01

Instant Vitals Triage

Score 0–100 in seconds using BP, SpO2, Glucose, Temp + Pregnancy status based on WHO/NHM rules.

02

Live Doctor Dashboard

Highest-risk patients always float to the top in red with AI clinical summary and status tracking.

03

ASHA Sahayak AI

Bilingual (English + हिंदी) clinical guidance for danger signs, pregnancy, first aid and more.

KEY FEATURES

What happens in the next 60 seconds

Fast Patient Intake

Log name, vitals & red-flag symptoms in under 60 seconds. Simple form, minimal training.

Live Doctor Dashboard

Highest-risk patients float to the top in red. Real-time triage with status tracking.

Instant Alerts

High-risk notifications to ASHA workers and family the moment a case is created.

Rules + AI Risk Engine

Instant score (0–100) labelled Low / Medium / High with clear clinical reasons.

AI Clinical Summary

Natural-language explanation so doctors understand urgency and decide in seconds.

ASHA Sahayak AI

Bilingual field assistant for danger signs, immunization, high-risk pregnancy & first aid.

WHAT MAKES US DIFFERENT

Why this approach wins

Dimension	Existing Approach	ArogyaPulse
Accessibility	Limited	Works in low-connectivity
Automation	Fully manual	Instant deterministic triage
Real-time feedback	Almost none	Live risk + AI guidance
Language support	Mostly English	English + हिंदी
Time to action	Hours to days	Under 60 seconds

TECHNICAL ARCHITECTURE

Simple. Reliable. Field-ready.

Frontend

Lightweight Web UI

Mobile-friendly, low-literacy design

Backend

.NET 10 / ASP.NET Core

High-performance API

Database

SQLite

Zero-config, works offline

Intelligence

Rules + AI Engine

WHO/NHM clinical protocols

Languages

English + हिंदी

Bilingual AI guidance

IMPACT

Turning reactive care into proactive care

Earlier Intervention

Deterioration is caught before it becomes an emergency

Fewer Hospitalisations

Reduced burden on PHCs, families and the system

Empowered ASHAs

Real-time AI guidance builds confidence and care quality

Better Doctor Time

Only critical cases reach the top of the queue

Scalable Model

Simple, bilingual, field-ready system for rural India

Lives Protected

Every high-risk pulse is seen in time

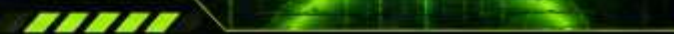


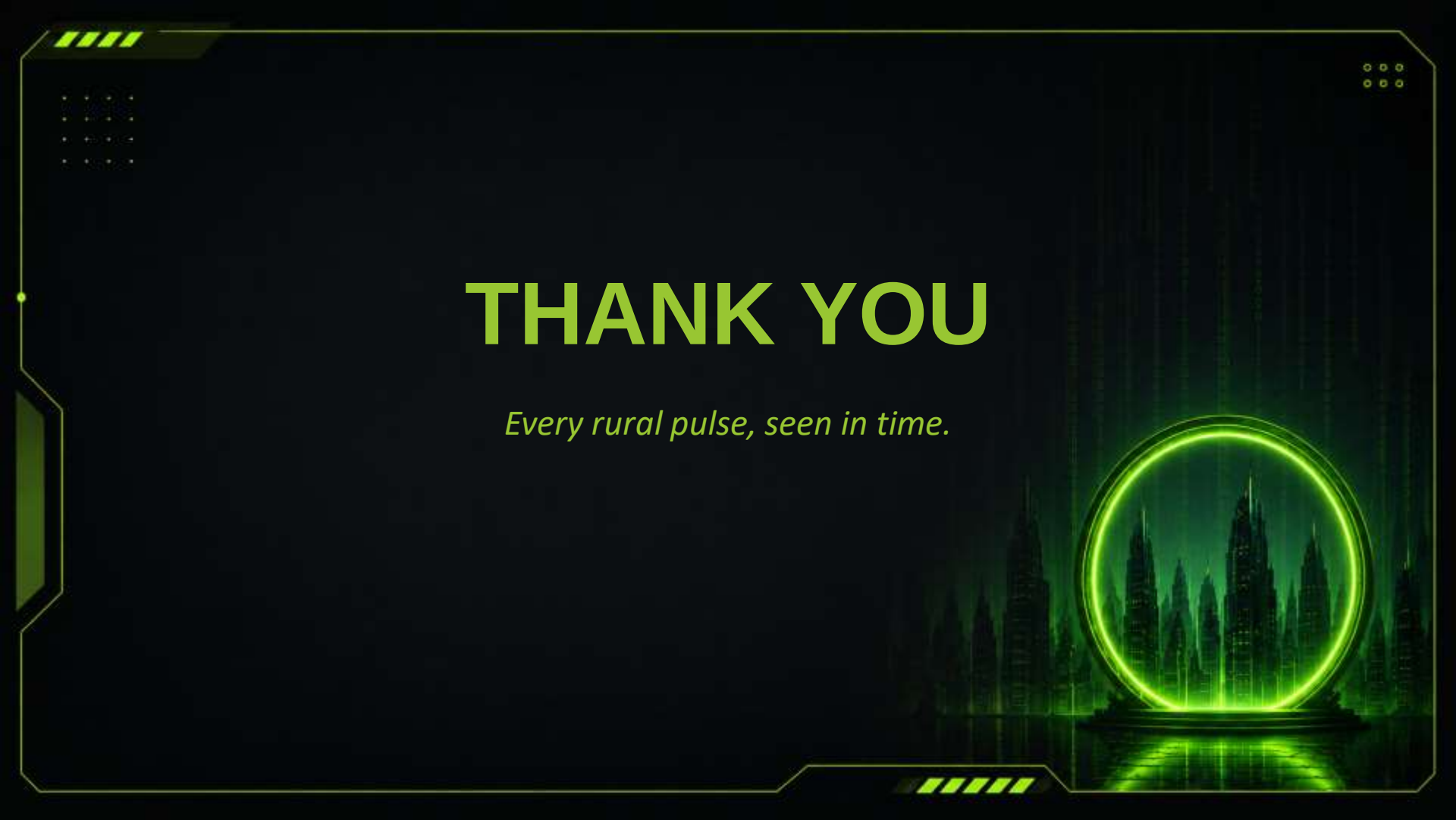
ArogyaPulse does not replace
doctors or ASHA workers.

**It amplifies them —
so every rural pulse is seen in time.**

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The background of the slide is a dark, futuristic cityscape at night. In the center-right, there is a large, glowing green circular portal or ring. Inside the ring, several tall, illuminated skyscrapers are visible. The city lights and the ring's glow reflect on a wet surface in the foreground. The entire scene is framed by a dark, metallic-looking border with various geometric shapes and glowing green lines. In the top-left corner, there are four slanted green bars. In the top-right corner, there are two small clusters of four green dots each. In the bottom-right corner, there are four slanted green bars. On the left side, there is a vertical green bar with a small white dot near the top.

THANK YOU

Every rural pulse, seen in time.